

The Job Churn Ladder

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Introduction

- Excess turnover, aka "churn" is the separations or hires in excess of the net employment change of the firm.
- Churn arises from poaching in search models feat. a job ladder (Burdett & Mortensen (1998), Postel-Vinay and Robin (2002), Moscarini & Postel-Vinay (2013), etc.)
- Implication of job ladder models is monotone ladder in churn (and wage) and churn/wage correlation at firm and worker level (Bagger & Lentz (2018), Sorkin (2018))

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 - Negative relationship between wages and turnover, and large dispersion in cross-section.

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 - Negative relationship between wages and turnover, and large dispersion in cross-section.
- We look for and find direct evidence of the “churn” ladder.
 - Consistent with models, workers move down a churn ladder as they move up a wage ladder.
- We fit the job ladder model to match our measures of the worker ladder and firm distribution.

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- **Separations to nonemployment** vary with firm wage, is key to matching the data.
- **Unobservable worker heterogeneity** in separation rates produces sorting over the ladder.
- Endogenous worker sorting accounts for **half** of the cross-sectional dispersion.
- Statistical decomposition over-attributes dispersion to firms.

Literature/Discussion

- Unobservable worker heterogeneity in attachment matters for understanding labor market dynamics, flows. Hall and Kudlyak (2019); Gregory et al. (2021), Ahn et al. (2023)
- Growing literature linking firm characteristics, dynamics with worker outcomes. Bilal, et al. (2022), Elsbay and Gottfries (2022), McCrary (2022), Tanaka et. al (2023), Ozkan et al. (2023)

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- Worker heterogeneity in separations is also key to understanding the firm distribution.
 - Composition of high and low separation workers changes over the wage distribution.
- Measuring Earnings Inequality. Abowd Kramarz, Margolis (1999), Borovickova and Macaluso (2023)
- Is the firm or the worker to blame for slippery job ladders? Jarosch (2023)

Data

- We use LEHD matched employer-employee data for 17 states from 1998-2013
- 15% random population sample, dropping firms < 10 workers
- Firm is a state employer identification number (SEIN)
- Job-to-Job transitions: overlapping quarters of employment, change in dominant job
- 1.2 million job transitions

Measurement

- Measure annual flows of firm offset 1 quarter from the transition period

- Define $H_{j,t} = \frac{\sum_{k=1}^4 \mathbb{H}_{j,t-k}}{\frac{1}{2}(L_{j,t} + L_{j,t-4})}$ and $S_{j,t} = \frac{\sum_{k=1}^4 S_{j,t-k}}{\frac{1}{2}(L_{j,t} + L_{j,t-4})}$

- Net Employment Growth Rate:

$$\Delta L_{j,t} = H_{j,t} - S_{j,t}$$

- Excess Accession (Churn) Rate:

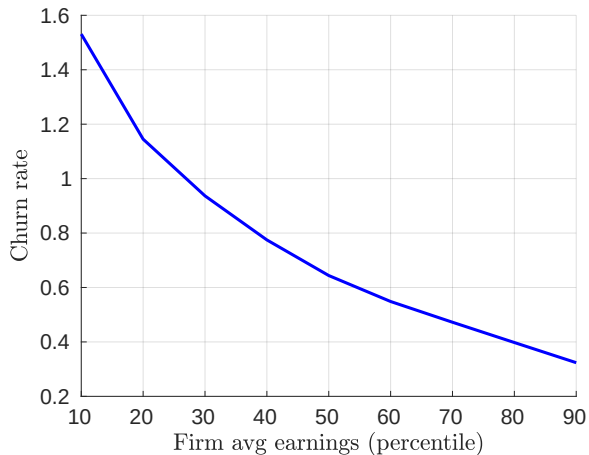
$$EAR_{j,t} = H_{j,t} - \max(\Delta L_{j,t}, 0) = \min(H_{j,t}, S_{j,t})$$

- Firms' Average Wage

$EW_{j,t}$: mean of employees' quarterly earnings

Churn and average wages

Firm Distribution



Measuring the job ladder

Excess Accessions Rate (Churn)

- Index jobs i within continuous employment spell j

-

$$EAR_{ij} = \beta_0 + \sum_{i=1}^N \beta_i \mathbb{I}_{job_i} + \gamma_j + \varepsilon$$

- Control for heterogeneity across workers and across employment spells with γ_j .
- We can sum coefficients and plot the change at each job index.

Measuring the job ladder

Firm Average Wage

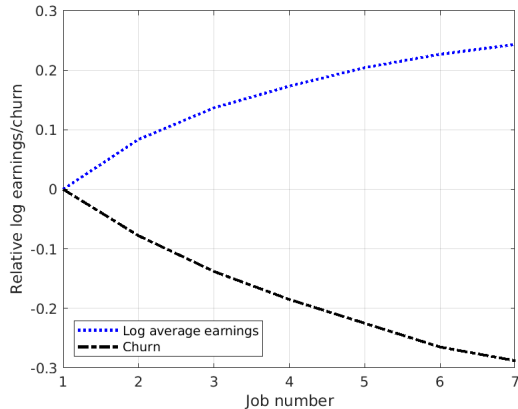
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The job ladder



Estimated with emp. spell FEs

► Estimated without fixed effects

The churn ladder in a search model

Environment: Burdett Mortensen (1998)

- Measure 1 of workers, pos. measure of firms of productivity z
- Random search in unemployment λ_u and on the job λ_e
- Exogenous separation into nonemployment δ
- Wage posting (constant, with commitment) and unemployment benefit b
- Offer distribution $F(w)$ results in realized wage distribution $G(w)$
- Firm size, conditional on w is $L(w)$

The churn ladder in a search model

Value Functions

Unemployed:

$$rV_u = b + \lambda_u \left[\int \max\{V_u, V_e(x)\} dF(x) - V_u \right]$$

Employed at wage w :

$$rV_e(w) = w + \lambda_e \left[\int \max\{V_e(w), V_e(x)\} dF(x) - V_e(w) \right] + \delta [V_u - V_e(w)]$$

The churn ladder in a search model

Flow balance without heterogeneity

The stationary distribution implies a flow balance between inflows and outflows

$$\underbrace{(\delta + \lambda_e(1 - F(w)))(1 - u)G(w)}_{\text{outflows}} = \underbrace{\lambda_u u F(w)}_{\text{inflows}}$$

- All flows are gross flows in steady state
- Separation rate is clearly decreasing in wages because $1 - F(w)$

The churn rate at wage w is:

$$c(w) = \delta + \lambda_e(1 - F(w))$$

Only $\lambda_e(1 - F(w))$ gives heterogeneity in churn.

- It will turn out, $\delta(w)$ varies with w .

Introducing worker, firm heterogeneity

- Permanent worker types (s), measure p has separation probability $\delta_s \in (0, \bar{\delta})$
- Types are unobservable to firm
- Reallocation shock λ_r to anywhere in the F distribution
- Firm-level separation rate elasticity with wage: $e^{\varepsilon_{\delta} w}$

The churn ladder in a search model

Value functions

Unemployed of type s :

$$rV_{u,s} = b + \lambda_u \left[\int \max\{V_{u,s}, V_{e,s}(x)\} dF(x) - V_{u,s} \right]$$

Employed of type s at wage w :

$$\begin{aligned} rV_{e,s}(w) = & w + \lambda_e \left[\int \max\{V_{e,s}(w), V_{e,s}(x)\} - V_{e,s}(w) \right] dF(x) \\ & + \lambda_r \left[\int V_{e,s}(\tilde{x}) - V_{e,s}(w) \right] dF(\tilde{x}) + e^{\varepsilon_{\delta} w} \delta_s \left[V_{u,s} - V_{e,s}(w) \right] \end{aligned}$$

The churn ladder in a search model

Adding heterogeneity

Type s has proportion p with separation rate $\delta_s \in (0, \bar{\delta})$

$$c(w) = \underbrace{e^{\varepsilon_{\delta} w} \int_0^{\bar{\delta}} \delta_s dD(s|w)}_{\delta(w)} + \lambda_r + \lambda_e(1 - F(w))$$

Two sources of heterogeneity:

- $e^{\varepsilon_{\delta} w}$ from firms
- $\int_0^{\bar{\delta}} \delta_s dD(s|w)$ from composition

The churn ladder in a search model

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The flow balance becomes

$$\int_{\underline{w}}^{\bar{w}} \int_0^{\bar{\delta}} (e^{\varepsilon_{\delta} w} \delta_s + \lambda_r(1 - F(w)) + \lambda_e(1 - F(w)))(1 - u)h(w, s) ds dw =$$
$$\lambda_u u F(w) + F(w) \int_{\underline{w}}^{\bar{w}} \int_0^{\bar{\delta}} \lambda_r(1 - u)h(w, s) ds dw$$

Exact identification and backing out $F(w)$

- With discrete wages (W levels) and 2 types of workers, exactly identified
- Observe $\delta(w)$ and $g(w)$. Defines a system of $2 \times W$ equations:

$$\delta(w_i) = \left(\sum_{j=1}^{\Delta} \delta_j \frac{h(w_i, \delta_j)}{g(w_i)} \right) e^{\varepsilon_{\delta} w_i}$$

$$g(w_i) = \sum_{j=1}^{\Delta} h(w_i, \delta_j)$$

implies the *joint* density $h(w_i, \delta_j)$.

- Discretized flow balance equation to solve for F , given h :

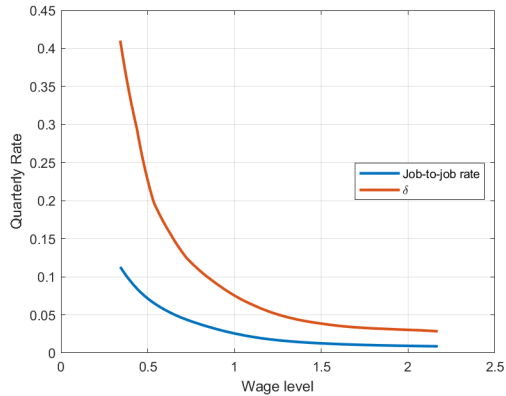
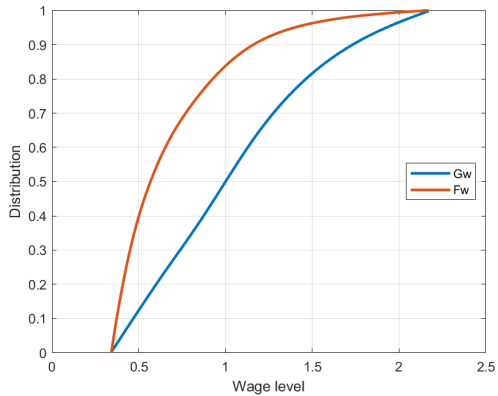
$$F(w_k) = \frac{\sum_{i=1}^k \sum_{j=1}^{\Delta} (\delta_j e^{\varepsilon_{\delta} w_i} + \lambda_r + \lambda_e)(1-u)h(w_i, \delta_j)}{(\lambda_r + \lambda_e) \sum_{i=1}^k \sum_{j=1}^{\Delta} (1-u)h(w_i, \delta_j) + \lambda_u \sum_{j=1}^{\Delta} u_j p_j + \sum_{i=k+1}^W \sum_{j=1}^{\Delta} \lambda_r (1-u)h(w_i, \delta_j)}$$

Calibration

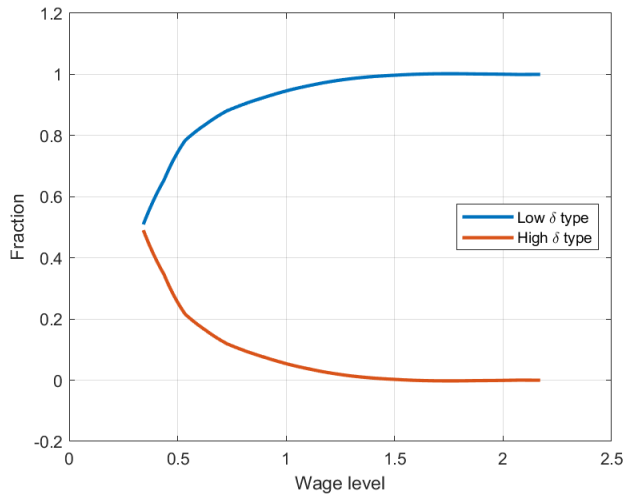
Take normalized $G(w)$ distribution from data - emp. weighted firm avg wage
Discretize wages and 2 worker types, proportion p_1 and p_2 with δ_1 and δ_2

Parameter	Value	Target
p_1	0.8	Ahn et al. (2023)
p_2	0.2	
λ_e	0.044	Quarterly EE rate 0.035 (J2J)
λ_u	0.084	Quarterly NE rate 0.084
λ_r	0.0175	est. - slope of worker ladder
$\varepsilon_{\delta w}$	-0.4053	est. - firm churn/wage dist.
δ_1	0.028	est. - $\min \delta(w)$ & $\delta(w) / J2J(w)$
δ_2	0.37	

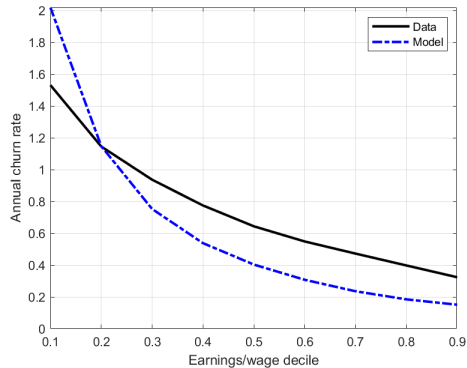
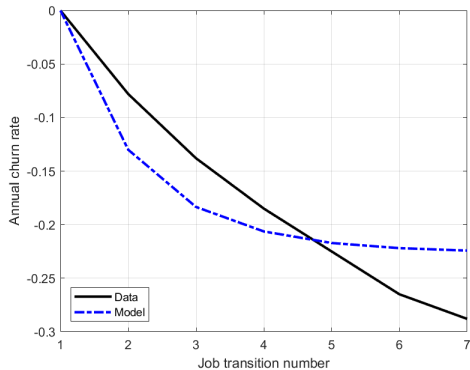
Calibrated churn ladder: Heterogeneous types



Calibrated churn ladder: Sorting over the job ladder

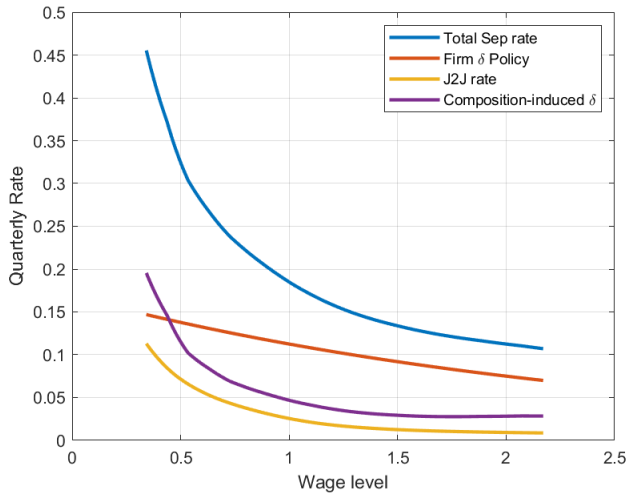


Calibrated churn ladder: With worker heterogeneity



► Const. Fraction of Seps.

Calibrated churn ladder: With worker heterogeneity

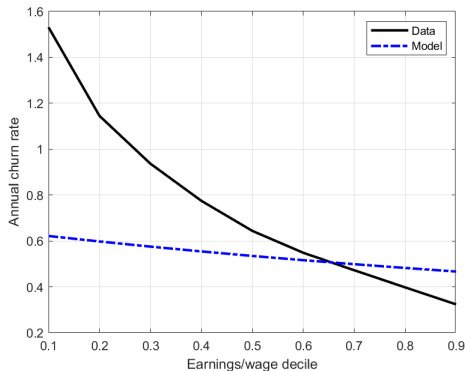
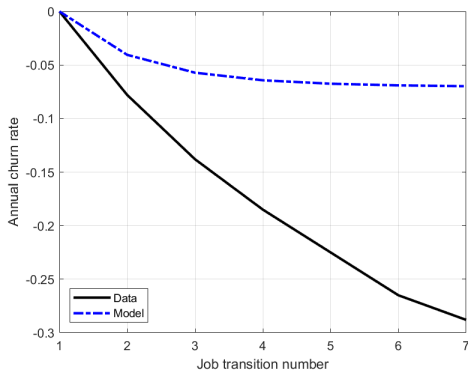


Counterfactuals

- What goes wrong with only OTJ search/Poaching?
- What if there's no worker sorting and firm policy drives everything?
- AKM with separation rates biases towards firms

Calibrated churn ladder: Only J2J

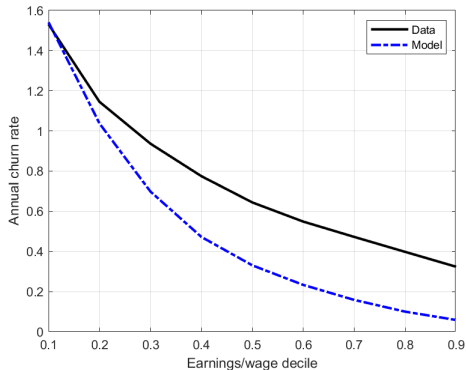
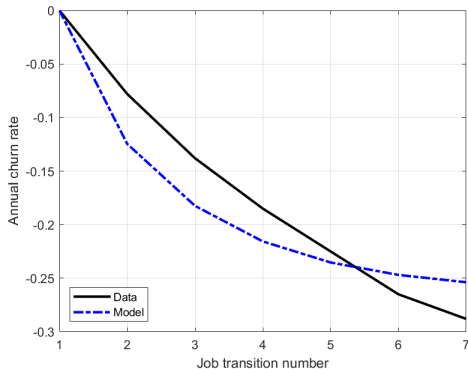
- Set $\epsilon_\delta = 0$ and $\delta_i = \bar{\delta}$, i.e. Burdett Mortensen (1998)
- $c(w) = \bar{\delta} + \lambda_e(1 - F(w))$ gets all heterogeneity from $F(w)$



- Considerably less decline in churn over job transitions and less dispersion at firm level.

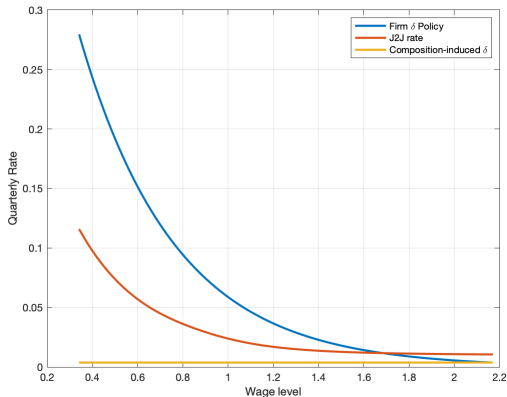
Firm Policy world: firm elasticity but no worker heterogeneity

- Set $\delta_i = \bar{\delta}$, i.e. Jarosch (2023) or Pinheiro & Visschers (2015)



- ϵ_{δ} can correctly identify the observed $\delta(w)$

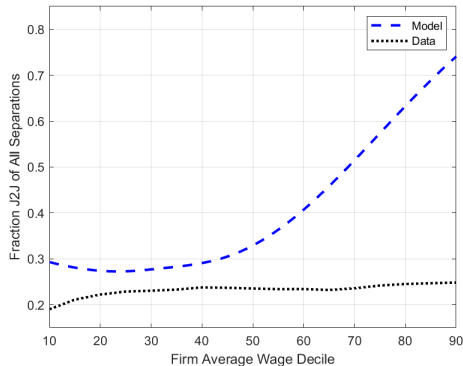
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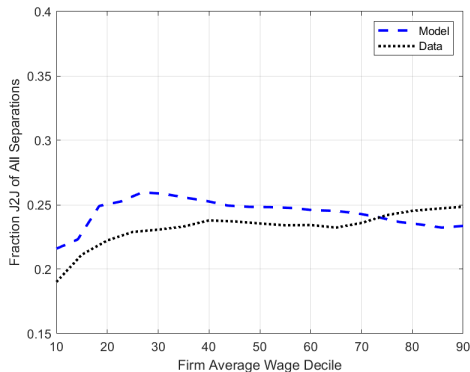
- There is no composition effect
- The firm policy has to pull $\delta(w)$ very steeply

Firm Policy world: firm elasticity but no worker heterogeneity

Firm Policy Only



Baseline



- $F(w)$ in this environment is different. For the same $G(w)$, $F(w)$ rises less steeply.
- More room for reallocation at the top of the job-ladder.

Abowd Kramarz Margolis-style decompositions

- Endogenous sorting puts high δ workers into high δ firms
- Using the empirical analog of separations, $d_{i,t} = \frac{1}{tenure_{j(i,t)}}$

$$\log d_{i,t} = \gamma_i + \phi_{j(i,t)} + v_{i,t}$$

because our “true” separation rate is log-additively separable this is correctly specified

- Estimation inflates the firms' role by 1.644 $\times$

	$var(\gamma_i)$	$var(\phi_j)$	Firms' fraction
Monte Carlo	0.2728	0.0128	4.49 %
True	0.6605	0.0186	2.73 %

The long-tail of duration gets assigned to low δ firms

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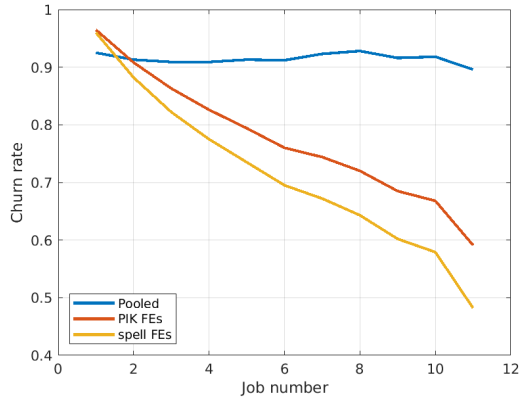
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- This is not small-t or limited-mobility bias

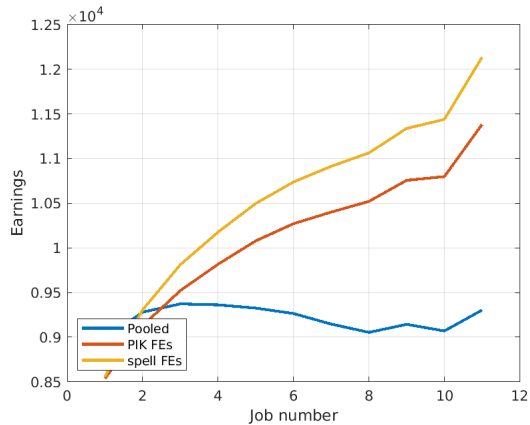
Conclusions

- Workers move down the churn ladder and up the wage ladder.
- Huge cross-sectional variation in churn in the data.
- Wage-turnover tradeoff from job ladder models is consistent with data.
- On-the-job search and separations to nonemployment both necessary to match churn.
- Worker heterogeneity in separation probability and selection key to matching data.

Fixed Effects

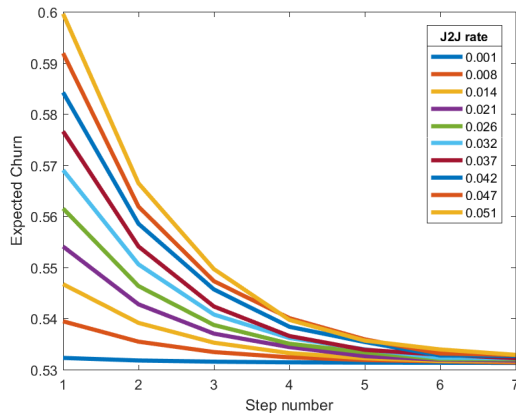


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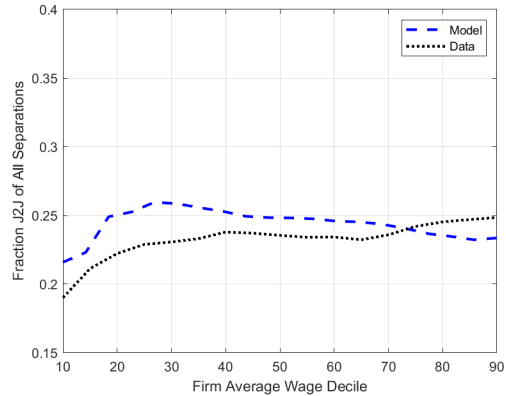
The Churn Ladder as λ_e changes

$$c(w) = \delta + \lambda_e(1 - F(w))$$



- No heterogeneity in δ

Fraction of all Separations to EE are Constant



Fraction of all Separations to EE are Constant (No Worker Sorting)

